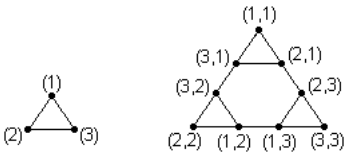
Sierpinski Gasket and Tower of Hanoi

In the discussion on *Sam Loyd's Fifteen puzzle (https://www.cut-the-knot.org/pythagoras/fifteen.shtml)* we introduced the notion of *puzzle graph (https://www.cut-the-knot.org/do_you_know/graphs2.shtml#puz(G))*. More generally the idea applies to every puzzle, game, or problem (not necessarily mathematical) that may be described by a set of parameters. With every move, parameter values change and the puzzle looks a little different. We are looking into a vector (an ordered set) of parameters whose values uniquely describe appearance (or *configuration*, or *state*) of the puzzle. To represent a puzzle graphically, associate various (legitimate) configurations of the puzzle with dots in some space (practically, it's a piece of paper or a computer screen we'll be working with). For every move from one configuration to another, connect the corresponding dots with an edge. If the move is not reversible, the edge must be *directed*. Otherwise, it's a plain curve connecting the two dots. To solve the puzzle, find dots corresponding to the initial and the target configuration and the path (of moves) from one to the other. If no such path exists the problem is unsolvable.

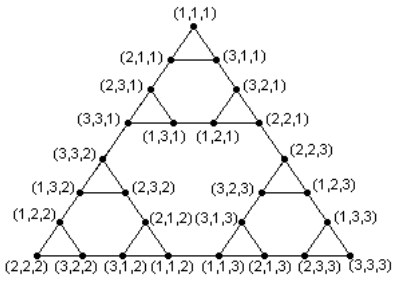
For a given puzzle G, *puz(G)* designates the associated puzzle graph. As was discovered by Ian Stewart, puz(*Tower of Hanoi (https://www.cut-the-knot.org/recurrence/hanoi.shtml)*) has a surprising relationship to the Sierpinski gasket (also known as the *Sierpinski triangle*) and, therefore, to Pascal's triangle.

In the *Tower of Hanoi (https://www.cut-the-knot.org/recurrence/hanoi.shtml)* puzzle, disks stacked on one peg are to be moved to another with, perhaps an intermediate stop at a third, auxiliary peg. Disks are all of different sizes and it's forbidden to place a bigger disk on top of a smaller one. Number the pegs 1,2, and 3. Every disk may be said to be at one of the pegs and, for a specific configuration, be associated with one of the numbers: 1, 2, 3. A puzzle configuration is completely defined by a list of such peg numbers associated with every disk. To avoid ambiguity the disks are ordered according to their sizes, with the smallest being the first.

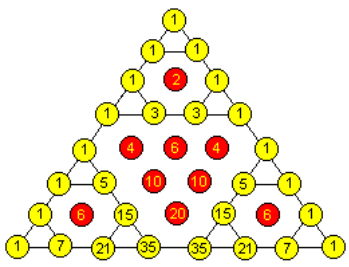


For a puzzle with a single disk, only three configurations are possible: (1), (2), (3), where, for example, (2) denotes the configuration in which the lone disk is located at the peg #2. For a puzzle with two disks, there are 9 possible configurations. (2, 2) means that both disks are located at the second peg. We'll tacitly focus on the legitimate disks positions. The only legitimate interpretation for (2, 2) is that the small disk is being on top, the bigger one at the bottom of the stack. Each of the 9 possible vectors corresponds to a single legitimate puzzle configuration. However, from a given node, not all configurations are reachable with a single move. From (2, 2) we may only get to (1, 2) and (3, 2) by moving the small disk from peg 2 to either peg 1 or peg 3. From (1, 2) we can get back to (2, 2) but also to (3, 2) and (1, 3). For instance, (1, 1) is off-limits as this would imply placing a bigger disk on top of a smaller one. The fun started when Stewart arranged the dots in a triangular pattern as above.

As Stewart observed, such pretty configuration can't be coincidence. Let puz(N) denote puz(Tower of Hanoi with N disks). Why puz(2) consists of three copies of puz(1)? Why there is a single edge from one copy of puz(1) to another? The explanation is not only easy, it's also generic: puz(N) always consists of three copies of puz(N - 1) with a single edge between any two copies of puz(N - 1)..



Indeed, place the largest disk on one of the three pegs and, for a while, forget about it. The remaining disks form a smaller puzzle. To three possible locations of the largest disk there correspond *three* "subpuzzles". The largest disk can only be moved with all the smaller ones stacked on a single peg (so that the third peg is free).



Well, the structure suspiciously resembles two famous triangles: Pascal's and *Sierpinski's (https://www.cut-the-knot.org/ctk/Sierpinski.shtml)*. We've *seen by induction (https://www.cut-the-knot.org/ctk/Sierpinski.shtml#induction)* that the Sierpinski gasket is Pascal's triangle modulo 2. puz(N) seems to look very much like the Sierpinski gasket with nearest dots connected by edges. In terms of Pascal's triangle, puz(N) combines nodes corresponding to *odd* entries of Pascal's triangle with the nodes 1 unit (whatever this may mean) apart linked by an edge. Since Pascal's triangle consists of *binomial coefficients*

(https://www.cut-the-knot.org/pigeonhole/sperner.shtml#binom) this suggests a question answered more than 100 years ago by Edouard Lucas (1842-1891): what is the *parity* of C(n, m)? In other words, when C(n, m) is odd and when is it even?

By now, we have established a solid base to be able to answer this question with little additional effort. First, let's present Pascal's triangle in its original, rectangular form. The apex goes to the upper left corner and the entries grow right and downward. The rows and columns are numbered starting with 0. We shall be detecting even entries in a process modeled after the *trema removal algorithm (https://www.cut-the-knot.org/triangle/Tremas.shtml)*.

	0	1	2	3	4	5	6	7
0	1	1	1	1	1	1	1	1
1	1	2	3	4	5	6	7	8
2	1	3	6	10	15	21	28	36
3	1	4	10	20	35	56	84	120
4	1	5	15	35	70	126	210	330
5	1	6	21	56	126	252	462	792
6	1	7	28	84	210	462	924	1716
7	1	8	36	120	330	792	1716	3432

The key to understanding the diagram is the *binary representation (https://www.cut-the-knot.org/binary.shtml)* of the row and column indices. In the gray square (that looks like a rectangle), all indices (numbers, 4, 5, 6, and 7) share a power of 2, viz., $2^2 = 4$. This is the highest power of two that appears in this small table. In the red squares, the row and column indices all share the previous power of two, $2^1 = 2$. In the yellow squares, all indices include the next power of 2, $2^0 = 1$, the smallest in this table. Note, that *even* entries fall inside one of the colored squares whereas *odd* entries stay outside. Swivel the table 45° to the right. With all entries modulo 2, the result will be exactly that of the *applet automation (https://www.cut-the-knot.org/ctk/Sierpinski.shtml#applet)* that leads to the Sierpinski gasket. Denote an entry in the row n and column m as D(n, m). What the above argument demonstrates is that entries D(n, m) for which binary representations $(n)_2$ and $(m)_2$ have at least one 1 in the same position (counting right-to-left), are even. D(n, m) is odd iff $(n)_2$ and $(m)_2$ never have 1s in the same binary position.

Now, this is immediate that D(n, m) = C(n + m, m) = C(n + m, n). Let's introduce another notation. For a given integer n, A_n is the set of non-zero binary positions in $(n)_2$. Positions are counted right-to-left starting with 0. For example, $11_{10} = 1011_2$, $17_{10} = 10001_2$, and $23_{10} = 10111_2$. Therefore, $A_{11} = \{0, 1, 3\}$, $A_{17} = \{0, 4\}$, and $A_{23} = \{0, 1, 2, 4\}$. As we just saw, D(n, m) is odd iff $A_n \cap A_m = \emptyset$ which is equivalent to $A_m \subset A_{n+m}$.

Now I state the *Lucas theorem*:

Theorem (E. Lucas)

C(n, m) is odd iff $A_m \subset A_n$.

For example, $A_{17} \subset A_{23}$ and, therefore, C(23, 17) is odd. Indeed, C(23, 17) = 100947. On the other hand, it's not true that $A_{11} \subset A_{23}$ and, therefore C(23, 11) is even. In fact, C(23, 11) = 1352978.

As another example, 2^n is the first index to have 1 in the n-th binary position. All its other digits are 0. Therefore, $C(2^n, m)$, n > 0, is always even, except the trivial case m = 0. This statement admits a *generalization (https://www.cut-the-knot.org/arithmetic/combinatorics/LucasTheorem.shtml)*, also known as *Lucas' theorem*: for p prime, a ≥ 1 and $0 < k < p^a$, $C(p^a, k) \equiv 0 \pmod p$.

The proof follows from the *algebraic definition (https://www.cut-the-knot.org/pigeonhole/sperner.shtml)* of the binomial coefficients:

$$C(n,m) = \frac{n!}{m! \cdot (n - m)!}$$

implying $k \cdot C(p^a, k) = p^a \cdot C(p^{a-1}, k-1)$ which shows that $C(p^a, k)$

is divisible by p whenever $0 < k < p^a$.

Denoting $N = p^a$, for p prime the *binomial theorem (https://www.cut-the-knot.org/Generalization/homo.shtml#binomial)* modulo p becomes

(1) $$(1 + x)^N = 1 + x^N \pmod p.$$

The most general theorem due to Lucas deals with arbitrary n, m, and a prime number p. The theorem deals with representations of n and m in *base p (https://www.cut-the-knot.org/blue/Modulo.shtml)*:

Theorem (E. Lucas)

Let $n = (a_x a_{x-1} \ldots a_0)_p$ and $m = (b_y b_{y-1} \ldots b_0)_p$. Let z = max(x, y) and pad the numbers on the left with zeros where necessary. Then

$$C(n, m) = C(a_z, b_z) \cdot C(a_{z-1}, b_{z-1}) \cdot \ldots \cdot C(a_0, b_0) \pmod p.$$

Proof

The assertion follows from *(1)*.

C(n, m) is the coefficient by x^m in the expansion of $(1 + x)^n$. But

$$\begin{aligned}(1 + x)^n &= (1 + x)^{a_z p^z} \cdot (1 + x)^{a_{z-1} p^{z-1}} \cdot \ldots \cdot (1 + x)^{a_0 p^0} \pmod p \\ &= [(1 + x)^{p^z}]^{a_z} \cdot [(1 + x)^{p^{z-1}}]^{a_{z-1}} \cdot \ldots \cdot [(1 + x)^{p^0}]^{a_0} \pmod p \\ &= (1 + x^{p^z})^{a_z} \cdot (1 + x^{p^{z-1}})^{a_{z-1}} \cdot \ldots \cdot (1 + x^{p^0})^{a_0} \pmod p.\end{aligned}$$

The power x^m, with $m = (b_z b_{z-1} \ldots b_0)_p$, comes up in this expansion only as a product of $x^{b_z p^z}$ from the first term, $x^{b_{z-1} p^{z-1}}$ from the second and so on, thus proving Lucas' theorem in its most general form.

Remark

While constructing puz(N) with N growing, we might want to keep the total length of one side of the triangle constant. It can be shown that thus defined sequence of point sets, puz(N), indeed converges to the Sierpinski gasket. Convergence is established in the framework of metric spaces with the *Hausdorff distance (https://www.cut-the-knot.org/do_you_know/Hausdorff.shtml)*.

References

1. I. Aharoni, *From Tower of Hanoi to Pascal's Triangle*, Alefefes, #4, 1996 (in Hebrew)
2. O. Ore, *Graphs and Their Uses (https://www.amazon.com/exec/obidos/ISBN=0883856352/ctksoftwareincA/)*, MAA, 1990
3. V. H. Moll, *Numbers and Functions: From a Classical-Experimental Mathematician's Point of View (https://www.amazon.com/exec/obidos/ISBN=0821887955/ctksoftwareincA/)*, AMS, 2012
4. I. Stewart, *Another Fine Math You've Got Me Into (https://www.amazon.com/exec/obidos/ISBN=0716723417/ctksoftwareincA/)*, W.H.Freeman & Co, 1992
5. I. Stewart, *Game, Set and Math (https://www.amazon.com/exec/obidos/ISBN=0140132376/ctksoftwareincA/)*, Penguin Books, 1989
6. D. Wells, *You Are a Mathematician (https://www.amazon.com/exec/obidos/ISBN=0471180777/ctksoftwareincA/)*, John Wiley & Sons, 1995

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